Old Exam Questions Reinforcement Learning

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How this file is organized

This file is separate from the course summary and is meant as a clean practice set. Exact duplicates were removed. When essentially the same question appeared in multiple old exams, that is marked with *Asked multiple times.* inside the question box. The wording was lightly normalized where the scans were messy, but the mathematical content and answer choices were kept intact.

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Introduction & Basics

Question 1: Match Learning Paradigms

Match the following definitions to the concept.

- **Definition 1:** Learn a function that maps inputs (features) to outputs (targets), based on a training set consisting of example input-output pairs.
- **Definition 2:** Learn how agents should take actions in an environment in order to maximize a reward.
- **Definition 3:** Learn patterns and find structure from unlabeled data.

Concepts: Reinforcement Learning, Supervised Learning, Unsupervised Learning.

Definition 1 → _____ Definition 2 → _____ Definition 3 → _____

Question 2: Unsupervised Learning Definition

Unsupervised learning considers the problem of goal-directed agents interacting with an uncertain environment.

☐ True

☐ False

Question 3: RL Use Case

Which of these applications is a good use case for reinforcement learning?

☐ A software to identify which pictures show a cat, a dog, or a hamster in a photo library

☐ A computer program that plays chess

☐ Clustering a dataset of audio recordings to find similar songs

☐ An algorithm that identifies the different plants in your garden by taking a picture with your phone

Question 4: RL Components Match

Match each definition with the concept. Each concept should be used only once.

A. Defines the agent's way of behaving at a given time _____

B. A model that mimics the behavior of the environment _____

C. A numerical value that defines what are good and bad events for the agent. The environment provides this value at each time step. _____

D. Indicates long-term desirability of states and is defined as the sum of all rewards that an agent can expect to accumulate over the future _____

Concepts: reward signal, policy, model of the environment, value function.

Question 5: Environment, Agent, Policy Match

Match each definition with the concept.

A. Setting or "world" within which the agent operates and interacts. _____

B. System or entity that makes decisions and takes actions based on the given policy, interacting with the environment. _____

C. Mapping from the state of the environment to the actions to be taken by the agent in those states. _____

Concepts: environment; policy; agent.

Question 6: RL vs Supervised Learning

Reinforcement learning is similar to supervised learning in which they both require explicit supervision in the form of a training set of inputs (features) and output (targets).

- ☐ True
- ☐ False

Question 7: Exploration Definition

In the context of the exploration-exploitation trade-off, what is one possible interpretation of the term exploration?

- ☐ Always selecting actions randomly from a given probability distribution
- ☐ Taking actions that led to high rewards in the past
- ☐ Not always taking the action with the highest reward
- ☐ An agent must balance taking actions that have proven to be beneficial in the past and trying actions that have not been selected before

Question 8: SL/UL Exploration Trade-off

Both unsupervised and supervised learning need to deal with the trade-off between exploration and exploitation.

- ☐ True
- ☐ False

Multi-Armed Bandits

Question 9: Exploration-Exploitation Methods in MAB

Which of these methods explicitly (i.e., with a hyperparameter) balance exploration and exploitation in multi-armed bandits?

- ☐ Upper Confidence Bound (UCB) action selection.
- ☐ Epsilon-greedy action selection
- ☐ Greedy action selection.

☐ The Gradient Bandit Algorithm

Question 10: Epsilon-Greedy Probability

In epsilon greedy action selection, for the case of 10 actions and $\epsilon = 0.7$, what is the probability that the greedy action is selected?

☐ 0.37

☐ 0.7

☐ 0.03

☐ 0.1

Question 11: ISAM Value Update

The arm a of a slot machine has a true mean reward of $q(a) = -1$. After $t = 10$ plays, your estimate of the mean reward is $Q_t(a) = -3$. The next play yields a reward of 2. What is the updated mean reward Q_{t+1} using the Incremental Sample-Averaging Method (ISAM) and a learning rate $\alpha = 0.1$?

☐ -2.5

☐ -2.9

☐ -3.1

☐ -0.5

Question 12: k-Armed Bandit Sample Averages

Consider a k-armed bandit problem with $k = 2$ actions (1 and 2). We use ϵ-greedy action selection and initial estimates $Q_1(1) = 0$ and $Q_1(2) = 0$. Suppose the initial sequence is:

- $t = 1$: $A_1 = 1, R_1 = 2$
- $t = 2$: $A_2 = 1, R_2 = -4$
- $t = 3$: $A_3 = 2, R_3 = -2$
- $t = 4$: $A_4 = 1, R_4 = -4$
- $t = 5$: $A_5 = 2, R_5 = -6$
- $t = 6$: $A_6 = 2, R_6 = -4$

Calculate the final value estimates $Q_7(1)$ and $Q_7(2)$ assuming sample-average updates.

$Q_7(1) =$ _____ $Q_7(2) =$ _____

Question 13: ISAM Step Size Role

The step size α in the action-value update equation of ISAM ...

- ☐ ...sets the probability of selecting a specific arm.
- ☐ ...adjusts the regret over time.
- ☐ ...controls how fast the algorithm explores new arms.
- ☐ ...determines the weight of new rewards in updating value estimates.

Question 14: Optimistic Initial Values Prior Knowledge

Optimistic initial values requires some prior knowledge about the reward distribution.

- ☐ True
- ☐ False

Question 15: Optimistic Initial Values Non-Stationary

The optimistic initial values method is more effective in non-stationary environments than stationary ones.

- ☐ True
- ☐ False

Question 16: UCB Uncertainty Decrease

In UCB, the uncertainty term for an action that is not selected can decrease.

- ☐ True
- ☐ False

Question 17: UCB Always Chooses Highest Value

In the UCB algorithm, actions with higher estimated value $Q(a)$ are always chosen.

- ☐ True
- ☐ False

Question 18: UCB Non-Stationary Suitability

Upper Confidence Bound (UCB) algorithms are well-suited for non-stationary problems.

- ☐ True
- ☐ False

Question 19: Exponential Recency-Weighted Averages

Exponential recency-weighted averages can be used to handle non-stationary problems.

- ☐ True
- ☐ False

Question 20: Gradient Bandit Model-Based

The Gradient Bandit algorithm is a form of model-based reinforcement learning and requires an approximation of the reward distribution.

- ☐ True
- ☐ False

Question 21: Gradient Bandit Action Selection

In the Gradient Bandit algorithm, actions are selected proportional to their action selection preference.

- ☐ True
- ☐ False

Question 22: Optimistic Initial Values Exploration Bias

Using optimistic initial values biases the agent toward initially exploring actions with higher estimated rewards.

- ☐ True
- ☐ False

Question 23: UCB Uncertainty Term

The UCB algorithm balances exploration and exploitation by adding an uncertainty term to the estimated value of each action.

- ☐ True
- ☐ False

Question 24: Non-Stationary Exploration Required

Non-stationary problems do not require balancing exploration and exploitation.

- ☐ True
- ☐ False

Question 25: Gradient Bandit Preference Updates

In the Gradient Bandit algorithm, action selection preferences are updated after each action.

- ☐ True
- ☐ False

Question 26: Epsilon Case First Occurrence

On which time steps did the ϵ-case definitely occur for the first time?

- ☐ 2
- ☐ 5

☐ 6

Question 27: Best Asymptotic Method

Which method will, on average, perform best in the long run (asymptotically) in terms of cumulative reward?

☐ ISAM, $\epsilon = 0.1$

☐ ISAM, $\epsilon = 0.01$

☐ ISAM, greedy

Question 28: Optimistic Initial Values Negative Rewards

If the rewards are strictly negative ($R < 0$), the Optimistic Initial Values method benefits from low initial value estimates ($Q_1 \ll 0$).

☐ True

☐ False

Question 29: Optimistic Initial Values Non-Stationary Tracking

Optimistic Initial Value methods are effective in tracking non-stationary k-armed bandit problems.

☐ True

☐ False

Question 30: UCB with c=0

UCB with $c = 0$ corresponds exactly to the greedy action selection method.

☐ True

☐ False

Question 31: UCB Uses Epsilon-Greedy

UCB relies on ϵ-greedy action selection for exploration.

☐ True

☐ False

Question 32: UCB Tries Each Action Once

UCB initially selects each as-yet-untried action exactly once.

☐ True

☐ False

Question 33: Gradient Bandit Random Exploration

Gradient Bandit algorithms do not rely on random action selection for exploration.

☐ True

☐ False

Question 34: Action Preferences vs Expected Reward

Numerical action selection preferences cannot be translated to estimates of the expected reward.

☐ True

☐ False

Question 35: Gradient Bandit Q-Value Baseline

The gradient bandit algorithm can be improved by using action value estimates $Q(a)$ as the baseline in the update rule instead of $\bar{R}_t$.

☐ True

☐ False

Finite Markov Decision Processes

Question 36: Stochastic Process Concepts Match

Match each definition to the concept. Each concept should only be used once.

A. A collection of random variables $\{S_t \mid t \in T\}$. The variables take values from a state space S, and are indexed by some ordered index set T. _____

B. Property according to which a stochastic process can be classified as discrete-space or continuous-space. _____

C. The future is independent of the past given the present, i.e., $p(S_{t+1} \mid S_t, S_{t-1}, \ldots) = p(S_{t+1} \mid S_t)$. _____

Concepts: Markov property; Stochastic Process; Cardinality of state space.

Question 37: MDP Equations Match

Match the following definitions/equations about Markov decision processes (MDPs) with the concept.

- $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$ _____
- $G_t = R_{t+1} + R_{t+2} + \cdots + R_T$ _____
- $p(s', r|s, a) = \Pr\{S_t = s', R_t = r|S_{t-1} = s, A_{t-1} = a\}$ _____

Concepts: Discounted return; Return (undiscounted); Dynamics of the MDP.

Question 38: Discounted Return Calculation

Suppose $\gamma = 1/2$ and the following sequence of rewards is received: $R_1 = -7, R_2 = 6, R_3 = 1, R_4 = 5, R_5 = 2$ with $T = 5$ (assume $R_t = 0$ for $t > T$). What are the returns $G_0, G_1, G_2, G_3, G_4, G_5$? (Hint: Work backwards.)

$G_5 =$ _____, $G_4 =$ _____, $G_3 =$ _____, $G_2 =$ _____, $G_1 =$ _____, $G_0 =$ _____

Question 39: Undiscounted Return Non-Episodic

It is better to use the undiscounted return (i.e., $\gamma = 1$) for non-episodic tasks.

- ☐ True
- ☐ False

Question 40: MDP Action Influences Next State

In general, in a Markov decision process, the decision of an action a in state S_t does not influence what the next state S_{t+1} will be.

- ☐ True
- ☐ False

Question 41: Infinite Discounted Return

Consider a non-episodic task where there is a constant reward of $R = 2$. What would be the discounted return after an infinite number of time steps, if the discount factor is $\gamma = 0.75$?

$G_t =$ _____

Dynamic Programming

Question 42: Policy Evaluation Computes Optimal Policy

Policy evaluation computes the optimal policy directly.

- ☐ True
- ☐ False

Question 43: Two-Array Policy Evaluation Update Order

When using the two-array version of the policy evaluation algorithm, the order in which states are updated can affect the speed of convergence.

- ☐ True
- ☐ False

Question 44: Greedy Policy in Policy Improvement

In the context of policy improvement, 'greedy policy' refers to a policy that explores actions randomly to gather more information.

- ☐ True
- ☐ False

Question 45: One Policy Always Better

Given two policies, one of them is guaranteed to be better than the other.

- ☐ True
- ☐ False

Question 46: Optimal Policy Greedy w.r.t. Value

The optimal policy is found if the policy is greedy with respect to its own value function.

- ☐ True
- ☐ False

Question 47: Policy Iteration Finite State Space Only

The policy iteration algorithm introduced in the lecture can be used for both finite and infinite state spaces.

- ☐ True
- ☐ False

Question 48: Value Iteration Greedy Policy

Value iteration indirectly evaluates a greedy policy with respect to the current value function estimate.

- ☐ True
- ☐ False

Question 49: Value Iteration Alternates Eval and Improvement

Value iteration alternates between policy evaluation and policy improvement.

- ☐ True
- ☐ False

Question 50: Policy Evaluation Convergence Guarantee

Policy evaluation is guaranteed to converge to the value function of the given policy.

- ☐ True
- ☐ False

Question 51: Policy Improvement Strictly Better

Policy improvement in dynamic programming always leads to a strictly better policy.

- ☐ True
- ☐ False

Question 52: Policy Iteration Not Guaranteed Finite

In policy iteration, an optimal policy is not guaranteed to be found after a finite number of steps.

- ☐ True
- ☐ False

Question 53: Policy Iteration Guaranteed Finite

*Asked multiple times.*Policy Iteration is guaranteed to find an optimal policy after a finite number of steps.

- ☐ True
- ☐ False

Question 54: Iterative Policy Evaluation Optimal Policy

Iterative Policy Evaluation is guaranteed to find the optimal policy after a finite number of steps.

- ☐ True
- ☐ False

Question 55: Value Iteration Finite Convergence

Value iteration converges to the optimal value function after a finite number of iterations.

- ☐ True
- ☐ False

Question 56: Value Iteration Stochastic Unique Optimum

If the environment is stochastic, value iteration always converges to a unique optimal value function.

- ☐ True
- ☐ False

Question 57: Value Iteration Bellman Optimality Update

Value iteration uses a Bellman-optimality-equation-like update rule.

☐ True

☐ False

Question 58: Iterative Policy Evaluation Bootstrapping

Iterative Policy Evaluation uses bootstrapping to estimate a state's value.

☐ True

☐ False

Question 59: Iterative Policy Evaluation Successor States

Iterative Policy Evaluation uses the value estimates of all immediate successor states to update the current state's value.

☐ True

☐ False

Question 60: Optimal Policy via Linear Equations

Given an arbitrary policy, the optimal policy can be computed directly by solving a system of linear equations using the Bellman equation.

☐ True

☐ False

Question 61: Policy Improvement Small vs Large Step

Consider a state 0 where the agent can choose between taking a **small** or a **large** step to the right.

- The **small step** takes the agent into state 1 or state 3 with probability 0.5 each.
- The **large step** takes the agent into state 2 (prob 0.5), state 4 (prob 0.25), or state 5 (prob 0.25).

Given the following values: $V(1) = 3, V(2) = 5, V(3) = -1, V(4) = -3, V(5) = -7$, reward $R = -1$ for all transitions, and $\gamma = 1$.

1. Calculate $Q(\text{small})$: _____

2. Calculate $Q(\text{large})$: _____
3. Which action does policy improvement select? _____

Question 62: DP Iterative Policy Evaluation Grid

Consider a grid world with states 1, 2, and 3, plus a terminal state (grey). Reward $= -1$ per transition. Deterministic environment. Discount $\gamma = 0.5$.

Current values: $V(1) = 0, V(2) = 0, V(3) = 0$.

Perform one sweep of iterative policy evaluation (Two-array version).

- $V_{t+1}(1)$: _____
- $V_{t+1}(2)$: _____
- $V_{t+1}(3)$: _____

Question 63: Policy Improvement Grid Q-Values

Consider a grid world with the following current value estimates $V_t(s)$:

State 1 ($V = 4$)	State 2 ($V = 2$)	State 3 ($V = 0$)
	State 4 ($V = 8$)	

The agent receives a reward of $R = -1$ for each transition. The environment is stochastic: moves succeed with probability 0.5, or slide to a perpendicular cell with probability 0.25 each. $\gamma = 1.0$.

Calculate the q-values for State 2:

- $Q(2, \leftarrow)$: _____
- $Q(2, \rightarrow)$: _____

- $Q(2, \uparrow)$ (wall, assume stays put or slides): _____
- $Q(2, \downarrow)$: _____

Question 64: DP Grid Policy Improvement Action

Which action will policy improvement select for state 2 in the Dynamic Programming Iterative Policy Evaluation (Grid) problem?

☐ $\rightarrow$

☐ $\downarrow$

☐ $\uparrow$

☐ $\leftarrow$

Question 65: Policy Evaluation Two-Sweep Value

Consider a grid world where the environment gives a reward of -4 for every transition and the discount factor is 0.5. Assume the terminal state value is always 0. What is the estimated value of state 0 after **two sweeps** of policy evaluation? (Assume initial values are 0.)

_____

Monte Carlo Methods

Question 66: MC Control Methods Match

For each of the following statements, select the corresponding method from the list below:

1. **Statement A:** The probability of selecting the greedy action for the k-th episode is given by $\epsilon = 1/k$.
2. **Statement B:** The probability of selecting the greedy action is given by $1 - \epsilon$, with ϵ being a constant.
3. **Statement C:** Learn about target policy π from experience sampled from a behavior policy μ.

Methods:

☐ Off-policy learning

☐ On-policy MC control for epsilon soft policies

☐ Greedy in the limit with infinite exploration (GLIE) MC control

Statement A $\rightarrow$ _____

Statement B $\rightarrow$ _____

Statement C $\rightarrow$ _____

Question 67: MC Exploring Starts vs GLIE vs Off-Policy

For each of the following statements, select the corresponding method:

1. An ϵ-greedy method where the value of ϵ decreases every episode. _____
2. Use importance sampling to weight the return according to the similarity between the target (π) and behavior (μ) policies. _____
3. In this method, start each episode at random state-action pairs (all state-action pairs have non-zero probabilities of being at the start). _____

Concepts: GLIE; MC control with exploring starts; Off-policy learning.

Question 68: First-Visit MC Correlation

Value estimates in the first-visit Monte Carlo Policy estimation algorithm are correlated with each other.

☐ True

☐ False

Question 69: MC Control Estimates State Values

Monte Carlo control methods compute an estimate of the state value function v_π.

☐ True

☐ False

Question 70: MC Convergence Speed

In practice, Monte Carlo methods for value estimation require just a few episodes to converge to the value function.

☐ True

☐ False

Question 71: MC Requires Dynamics Knowledge

Monte Carlo methods require explicit knowledge of the dynamics of the problem ($p(s', r|s, a)$).

☐ True

☐ False

Question 72: Fish Lake First-Visit MC

Following policy π, we sample the following 4 episodes in a "Fish Lake" grid world (actions: left, right; rewards: 0, 1, -1):

- **Ep 1:** $S_0 = b, A_0 = \text{left}, R_1 = 0, S_1 = b, A_1 = \text{left}, R_2 = 0, S_2 = b, A_2 = \text{right}, R_3 = 1, S_3 = c$
- **Ep 2:** $S_0 = b, A_0 = \text{left}, R_1 = 0, S_1 = b, A_1 = \text{left}, R_2 = 0, S_2 = b, A_2 = \text{left}, R_3 = -1, S_3 = a$
- **Ep 3:** $S_0 = b, A_0 = \text{right}, R_1 = 0, S_1 = b, A_1 = \text{right}, R_2 = 0, S_2 = b, A_2 = \text{right}, R_3 = 1, S_3 = c$
- **Ep 4:** $S_0 = b, A_0 = \text{left}, R_1 = 0, S_1 = b, A_1 = \text{right}, R_2 = 0, S_2 = b, A_2 = \text{right}, R_3 =$

$1, S_3 = c$

Using **First Visit Monte Carlo Policy Evaluation** ($\gamma = 1$):

1. Estimate $q_\pi(b, \text{left})$: _____
2. Estimate $q_\pi(b, \text{right})$: _____

Question 73: MC Model-Free

Monte Carlo methods are model-free.

☐ True

☐ False

Question 74: Every-Visit MC IID Samples

Samples from the every-visit Monte Carlo Policy estimation algorithm are independent and identically distributed.

☐ True

☐ False

Temporal Difference Learning

Question 75: TD Policy Evaluation Grid S0-Start

Consider a grid world where the agent follows a deterministic policy π:

- From S_0, go Right to S_1.
- From S_1, go Up to S_2.
- From S_2, go Left to S_T (Terminal).

Parameters: $\gamma = 1$, Reward $R = -1$ per step, $\alpha = 1$. Initial values $V(S) = 0$.

Perform TD(0) updates for one episode: $S_0 \to S_1 \to S_2 \to S_T$.

Calculate the updated values (round to 3 decimal places):

- $V(S_1)$ after the step $S_0 \to S_1$: _____
- $V(S_2)$ after the step $S_1 \to S_2$: _____

Question 76: Off-Policy SARSA Statements

Assuming the MDP above (TD Policy Evaluation S0-Start), we use an off-policy version of SARSA, using the deterministic target policy π as indicated by the arrows, and a non-deterministic behavior policy μ to explore. $Q(S_t, A_t)$ is initialized as 0 for all states and actions. The agent always starts the episode in state S_0. Which of the following statements are true?

- ☐ Following the deterministic policy $\pi(a|S)$, some state action pairs are never explored.
- ☐ $Q(S_t, A_t)$ for actions A_t chosen by the deterministic policy converges to the TD(0) policy evaluation solution $V(S_t)$ computed before.
- ☐ This version of SARSA has no optimality guarantees because the discount factor γ is larger than 0.
- ☐ This off-policy version of SARSA would be identical to Q-learning.

Question 77: Q-Learning Off-Policy No IS

Q-Learning is an off-policy method that does not require importance sampling.

- ☐ True
- ☐ False

Question 78: TD Advantage over MC

What is an advantage of TD learning over Monte Carlo methods?

- ☐ In practice, TD learning converges faster because it doesn't need to wait for an episode to end.
- ☐ It always gives a more accurate estimate.
- ☐ TD learning requires the dynamics of the environment, while MC methods are model-free.
- ☐ It works better in deterministic environments.

Question 79: Expected SARSA Equals Q-Learning

Expected SARSA and Q-Learning can be the same for some choice of policy.

- ☐ True

☐ False

Question 80: TD Non-Episodic Tasks

What is the primary reason TD learning can handle non-episodic tasks better than Monte Carlo methods?

☐ TD learning does not require a discount factor.

☐ TD learning is model-free, while Monte Carlo is model-based.

☐ TD learning updates based on partial episodes, allowing for continuous learning.

☐ TD learning can compute exact state values in one step.

Question 81: SARSA Second Action Same Policy

SARSA requires choosing a second action in the second state with the same policy as the first.

☐ True

☐ False

Question 82: SARSA vs Expected SARSA Direction

SARSA moves deterministically in the direction Expected-SARSA is moving in expectation.

☐ True

☐ False

Question 83: N-Step TD Equivalent to Exhaustive Search

TD methods can take many steps at the same time, effectively interpolating between TD(0) and Exhaustive Search methods.

☐ True

☐ False

Question 84: N-Step Intermediate Performance

N-step methods somewhere in between TD(0) and TD(1) will often perform better than either extreme.

☐ True

☐ False

Question 85: TD(lambda) Weighted Average

The TD(λ) algorithm computes a weighted average n-step TD backup target.

☐ True

☐ False

Question 86: Eligibility Traces Forward View

Eligibility Traces implement a forward view of TD(λ) methods.

- ☐ True
- ☐ False

Question 87: TD(0) Update Rule

Which of the following equations represents the TD(0) update rule for policy evaluation?

- ☐ $V(s) \leftarrow V(s) + \alpha[G_t - V(s)]$
- ☐ $V(s) \leftarrow V(s) + \alpha[R + \gamma V(s') - V(s)]$
- ☐ $Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

Question 88: TD(0) Discount Factor Zero

In policy evaluation using TD(0), what happens if the discount factor (γ) is set to 0?

- ☐ The agent considers only immediate rewards.
- ☐ The agent evaluates the policy using the average reward per time step.
- ☐ The learning process diverges.
- ☐ The value function estimates future rewards up to an infinite horizon.

Question 89: TD(0) Bias vs MC

TD(0) generally is less biased than Monte Carlo methods.

- ☐ True
- ☐ False

Question 90: TD(0) vs MC Key Difference

What is the key difference between TD(0) and Monte Carlo methods in policy evaluation?

- ☐ TD(0) only uses undiscounted returns, while Monte Carlo uses both undiscounted and discounted returns.
- ☐ TD(0) requires a model of the environment; Monte Carlo does not.
- ☐ TD(0) updates after observing a full episode, while Monte Carlo updates after every step.
- ☐ TD(0) updates after every step, while Monte Carlo updates after observing a full episode.

Question 91: N-Step TD Equivalent to Dynamic Programming

TD methods can take many steps at the same time. If the number of steps is greater than the episode length, TD methods are equivalent to Dynamic Programming.

- ☐ True
- ☐ False

Question 92: N-Step TD Incomplete Episodes

N-step TD methods have to wait n time steps before making an update but can update from incomplete episodes.

- ☐ True
- ☐ False

Question 93: TD Policy Evaluation S3-Start

Consider a grid world with states 1, 2, 3 and Terminal. Reward $= -1$. Deterministic. $\gamma = 0.5, \alpha = 0.5$.

Current values: $V(1) = 0, V(2) = 0, V(3) = 0$.

Episode: Start at $S_3 \to$ Right $\to S_2 \to$ Up $\to S_1 \to$ Left $\to S_T$.

Calculate the updates:

- $V(S_3)$ after first step: _____
- $V(S_2)$ after second step: _____

Question 94: TD(0) Convergence Settings

Assuming the MDP from the "TD Policy Evaluation (S3 start)" question, under which settings for the step size α and the discount rate γ does the TD(0) estimate converge to the same estimate as the Dynamic Programming solution?

- ☐ $\alpha = 0.1, \gamma = 0.9$
- ☐ $\alpha = 1.0, \gamma = 0.5$
- ☐ $\alpha = 0.5, \gamma = 0.5$
- ☐ $\alpha = 0.5, \gamma = 0.9$

Question 95: SARSA on S3-Start Grid

Assuming the MDP from the "TD Policy Evaluation (S3 start)" question, we use the SARSA algorithm for TD(0) control while keeping the deterministic behavior policy. $Q(S_t, A_t)$ is initialized as in the figure for all actions. Which statements are true?

- ☐ This version of SARSA has no optimality guarantees because the discount factor γ is less than 1.0.
- ☐ This version of SARSA converges to the globally optimal policy.
- ☐ This version of SARSA has no optimality guarantees because some state action pairs are

never explored.

- ☐ $Q(S_t, A_t)$ for actions chosen by the deterministic policy converges to the TD(0) policy evaluation solution computed before.

Question 96: TD(0) Exact Same Sequence as DP

Under which settings for the step size α and the discount rate γ does the TD(0) policy evaluation produce the **exact same sequence** of estimates as the Dynamic Programming in the earlier section of the exam?

- ☐ $\alpha = 0.5, \gamma = 0.9$
- ☐ $\alpha = 0.5, \gamma = 1.0$
- ☐ $\alpha = 0.5, \gamma = 0.5$
- ☐ $\alpha = 1.0, \gamma = 0.5$

Question 97: Expected SARSA vs SARSA Non-Greedy

Expected SARSA and SARSA can be the same for some choice of non-greedy behavior policy.

- ☐ True
- ☐ False

Question 98: TD Requires Environment Model

TD methods require a model of the environment (i.e., the dynamics and reward function).

- ☐ True
- ☐ False

Question 99: TD(0) Bootstrapping

TD(0) uses bootstrapping (i.e., they update the value function using previous estimates of the value function).

- ☐ True
- ☐ False

Question 100: Batch MC vs Batch TD Maximum Likelihood

Both batch Monte Carlo learning and batch Temporal Difference learning converge to the maximum likelihood Markov model.

- ☐ True
- ☐ False

Question 101: N-Step TD Equivalent to MC

TD methods can take many steps at the same time. If the number of steps is greater than the episode length, TD methods are equivalent to Monte Carlo methods.

☐ True

☐ False

Question 102: TD(lambda) Semi-Gradient Convergence

TD(λ) targets can be used in semi-gradient methods without convergence guarantees.

☐ True

☐ False

Policy Gradient Methods

Question 103: Policy Gradient Log Probability

In the policy gradient theorem, the term $\nabla_\theta \log \pi_\theta(a|s)$ represents the direction in parameter space that increases the probability of selecting action a.

☐ True

☐ False

Question 104: Policy Gradient Deterministic Only

The policy gradient theorem applies only to deterministic policies.

☐ True

☐ False

Question 105: REINFORCE Deterministic Only

REINFORCE can only be used to learn deterministic policies.

☐ True

☐ False

Question 106: REINFORCE Episodic Only

REINFORCE is effective for episodic and non-episodic Markov decision processes.

☐ True

☐ False

Question 107: REINFORCE Baseline Reduces Variance

Using an appropriate baseline in REINFORCE with baseline reduces the variance of the gradient estimator.

☐ True

☐ False

Question 108: REINFORCE with Baseline Actor-Critic

The REINFORCE algorithm with baseline is considered an actor-critic method.

- ☐ True
- ☐ False

Question 109: Actor-Critic Variance Issues

Actor-critic methods suffer from the same variance issues as REINFORCE because they do not use baselines.

- ☐ True
- ☐ False

Question 110: Actor-Critic Critic Gradient

Actor-critic methods require the gradient of the critic to compute the policy update.

- ☐ True
- ☐ False

Question 111: Policy Gradient State Distribution

The policy gradient theorem states that the gradient of the expected reward can be expressed without the need to compute the gradient of the state distribution.

- ☐ True
- ☐ False

Question 112: REINFORCE Requires Value Function

REINFORCE requires a value function approximation to compute the policy update.

- ☐ True
- ☐ False

Question 113: REINFORCE On-Policy

The REINFORCE algorithm is an on-policy method, i.e., it requires trajectories generated by the current policy.

- ☐ True
- ☐ False

Question 114: REINFORCE Baseline Must Be Constant

The baseline in REINFORCE with baseline must be a constant value for the gradient estimate to remain unbiased.

- ☐ True
- ☐ False

Question 115: REINFORCE Baseline as Value Function

A common choice for the baseline in REINFORCE with baseline is an estimate of the value function $V(s)$.

- ☐ True
- ☐ False

Question 116: Actor-Critic Critic Learns State Value

The critic in actor-critic methods typically learns an approximation of the state value function $V(s)$.

- ☐ True
- ☐ False

Question 117: PPO Clipped Surrogate Objective

Proximal Policy Optimization (PPO) is an actor-critic method that uses a clipped surrogate objective to stabilize training.

- ☐ True
- ☐ False

Question 118: Policy-Based RL Uses Action Values

Policy-based RL methods rely on action values to select actions.

- ☐ True
- ☐ False

Question 119: Supervised Pretraining for Policy RL

Supervised pretraining with human-generated state-action pairs is an appropriate strategy to inject expert knowledge into a policy-based RL model.

- ☐ True
- ☐ False

Question 120: Policy-Based RL Exploration via Sampling

Policy-based RL methods trade-off exploration and exploitation by sampling from the parameterized policy.

- ☐ True
- ☐ False

Question 121: Performance Gradient State Distribution

The exact performance gradient $\nabla J(\Theta)$ depends on changes in the state distribution; a proportional quantity is often used as a substitute.

- ☐ True

☐ False

Question 122: REINFORCE with Baseline TD Method

The REINFORCE algorithm with baseline is considered a Temporal Difference learning method because it uses bootstrapping with state value estimates to estimate returns.

☐ True

☐ False

Question 123: REINFORCE with Baseline Non-Episodic

The REINFORCE algorithm with baseline can be used for non-episodic tasks.

☐ True

☐ False

Answer Key

Q1. Definition 1 → Supervised Learning; Definition 2 → Reinforcement Learning; Definition 3 → Unsupervised Learning

Q2. **False**

Q3. A computer program that plays chess

Q4. A → policy; B → model of the environment; C → reward signal; D → value function

Q5. A → environment; B → agent; C → policy

Q6. **False**

Q7. Not always taking the action with the highest reward

Q8. **False**

Q9. UCB and Epsilon-greedy (both correct)

Q10. 0.37

Q11. -2.5

Q12. $Q_7(1) = -2$; $Q_7(2) = -4$

Q13. ...determines the weight of new rewards in updating value estimates.

Q14. **True**

Q15. **False**

Q16. **False**

Q17. **False**

Q18. **False**

Q19. **True**

Q20. **False**

Q21. **True**

Q22. **True**

Q23. **False**

Q24. **True**

Q25. 5

Q26. ISAM, $\epsilon = 0.01$

Q27. **False**

Q28. **False**

Q29. **True**

Q30. **False**

Q31. **True**

Q32. **False**

Q33. A $\rightarrow$ Stochastic Process; B $\rightarrow$ Cardinality of state space; C $\rightarrow$ Markov property

Q34. First bullet $\rightarrow$ Discounted return; Second bullet $\rightarrow$ Return (undiscounted); Third bullet $\rightarrow$ Dynamics of the MDP

Q35. $G_5 = 0, \quad G_4 = 2, \quad G_3 = 6, \quad G_2 = 4, \quad G_1 = 8, \quad G_0 = -3$

Q36. **False**

Q37. **False**

Q38. $G_t = 8$

Q39. **False**

Q40. **False**

Q41. **False**

Q42. **False**

Q43. **True**

Q44. **False**

Q45. **True**

Q46. **False**

Q47. **True**

Q48. **False**

Q49. **False**

Q50. **True**

Q51. **False**

Q52. **False**

Q53. **True**

Q54. **True**

Q55. **True**

Q56. **False**

Q57. $Q(\text{small}) = 0; \quad Q(\text{large}) = -1;$ selects **small step**

Q58. Requires grid figure from slides — see problem setup (values depend on neighbor positions)

Q59. $\downarrow$

Q60. -6 (after two sweeps: first sweep gives -4, second sweep gives $-4 + 0.5 \times (-4) = -6$)

Q61. Statement A $\rightarrow$ GLIE MC control; Statement B $\rightarrow$ On-policy MC control for epsilon soft policies; Statement C $\rightarrow$ Off-policy learning

Q62. 1 $\rightarrow$ GLIE; 2 $\rightarrow$ Off-policy learning; 3 $\rightarrow$ MC control with exploring starts

Q63. **False**

Q64. **False**

Q65. **False**

Q66. **False**

Q67. $q_\pi(b, \text{left}) \approx 1/3 \approx 0.333; \quad q_\pi(b, \text{right}) = 1$

Q68. **True**

Q69. **False**

Q70. $V(S_1) = -1; \quad V(S_2) = -1$

Q71. Statements 1 and 2 are **True**; statements 3 and 4 are **False**

Q72. **True**

Q73. In practice, TD learning converges faster because it doesn't need to wait for an episode to end.

Q74. **True**

Q75. TD learning updates based on partial episodes, allowing for continuous learning.

Q76. **True**

Q77. **False**

Q78. **False**

Q79. **True**

Q80. **True**

Q81. **False**

Q82. $V(s) \leftarrow V(s) + \alpha[R + \gamma V(s') - V(s)]$

Q83. The agent considers only immediate rewards.

Q84. **False**

Q85. TD(0) updates after every step, while Monte Carlo updates after observing a full episode.

Q86. **False**

Q87. **True**

Q88. $V(S_3) = -0.5; \quad V(S_2) = -0.5$

Q89. $\alpha = 1.0, \gamma = 0.5$ and $\alpha = 0.5, \gamma = 0.5$ (both correct)

Q90. Statements 3 and 4 are **True**; statements 1 and 2 are **False**

Q91. $\alpha = 1.0, \gamma = 0.5$

Q92. **False**

Q93. **False**

Q94. **True**

Q95. **False**

Q96. True

Q97. True

Q98. True

Q99. False

Q100. False

Q101. False

Q102. True

Q103. False

Q104. False

Q105. False

Q106. True

Q107. False

Q108. True

Q109. False

Q110. True

Q111. True

Q112. True

Q113. False

Q114. True

Q115. True

Q116. True

Q117. False

Q118. False